ForceDelta-VLA: Supplementary Material

Anonymous Authors

I. Implementation Details

The temporal force-conditioned teacher is fully fine-tuned on eight NVIDIA A100 GPUs. All real-robot inference and runtime measurements use a single 48 GB NVIDIA RTX 4090D GPU. The teacher is trained for $50{,}000$ steps with AdamW at a peak learning rate of 2×10^{-5}. It predicts $H{=}50$ actions at 30 Hz from the latest 100 ms of the 100 Hz wrench stream, and flow-matching inference uses $N_{\text{flow}}{=}10$ steps.

After freezing the teacher, we train the rank-32 null-mode adapter for $10{,}000$ steps at a learning rate of 10^{-4}. The visual–language prefix is pooled into $M{=}16$ intent tokens. The residual policy comprises a causal force encoder and one pre-norm attention block with separate force and staleness heads. It predicts $K{=}5$ corrections and is trained for $30{,}000$ steps at a learning rate of $3{\times}10^{-4}$ under replayed asynchronous schedules.

The action representation and rotational residual composition are defined in Sec. III-A of the main paper. At deployment, force-conditioned and staleness corrections are bounded independently in the normalized pose-action space before composition with the reference action.